

TFScope: sequence-only prediction of transcription-factor binding specificity rivals structure-based methods

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Abstract

Predicting the DNA sequence preference of a transcription factor (TF) from its protein sequence alone remains a central problem in gene-regulatory genomics. State-of-the-art accuracy is achieved by structure-based models that require an experimentally determined or predicted protein–DNA complex, limiting their use for the large fraction of TFs and designed proteins that lack a reliable structure. Here we introduce **TFScope**, a sequence-only model that maps a TF DNA-binding domain (DBD) to a position weight matrix (PWM) using a *frozen* protein language model (ESM-2 650M) adapted with low-rank (LoRA) fine-tuning, a residue mixture-of-experts, and a contact-aware motif head. On a leakage-controlled benchmark in which held-out TFs share no training identity, a five-seed TFScope ensemble reaches a motif content correlation of 0.664 and is statistically indistinguishable from the structure-based method DeepPBS (mean gate-oracle r 0.657 vs 0.626; bootstrap $p = 0.30$), despite using no structure at inference. An independent structure predictor (Boltz-2) judges TFScope’s sequence-only consensus motifs to form protein–DNA complexes at least as confidently as DeepPBS’s (ipTM 0.957 vs 0.923; 32/41 TFs; Wilcoxon $p = 1 \times 10^{-5}$). Probing the frozen backbone shows that it linearly encodes DNA-contact residues (AUROC 0.95) and organizes DBD families (k-NN purity 0.83), explaining where the signal comes from; a variance decomposition attributes 59% of TFScope’s edge to genuine within-family specificity rather than family lookup. We further show directional single-mutation motif switching (MyoD1 E-box) and application to *de novo* designed DBDs, and we transparently characterize the model’s principal limitation — insensitivity to specificity-altering point mutations inherited from the frozen backbone. TFScope makes accurate specificity prediction available wherever only a protein sequence is known.

Introduction

Transcription factors read the genome by binding short DNA motifs, and a compact summary of that preference — the position weight matrix (PWM) — underlies much of regulatory-genomic analysis, from motif enrichment to variant interpretation^{1,2}. Experimental determination of PWMs by protein-binding microarrays or HT-SELEX is laborious³, and only a minority of the estimated ~1,600 human TFs, and essentially none of the rapidly growing space of designed DNA-binding proteins, have measured specificities. A predictive map from protein sequence to PWM would close this gap.

Two families of computational approaches exist. Structure-based models, most recently the geometric deep-learning method DeepPBS⁴, achieve strong accuracy by reading the physicochemical readout of an experimentally solved or predicted protein–DNA complex. Their accuracy is gated on structure availability and quality; for TFs without a co-crystal, and for designed proteins, a structure must first be predicted, adding cost and a second source of error. Sequence-only models sidestep structure but have historically lagged, and reported margins are frequently inflated by *TF leakage* — test proteins whose close homologs appear in training⁴.

Protein language models (PLMs) trained on evolutionary-scale sequence data have transformed sequence-only inference of protein structure and function^{5,6}. We hypothesized that a frozen PLM already contains the information needed to specify DNA-binding preference, and that a lightweight, structure-free adapter could extract it. Here we test this hypothesis with **TFScope** and show, under strict leakage control, that a sequence-only model matches a structure-based one — reaching accuracy parity without ever seeing a structure, while remaining applicable to any protein sequence.

Results

The TFScope model

TFScope maps a TF DNA-binding-domain (DBD) protein sequence to a $4 \times L$ PWM (Fig. 1[**TODO: architecture schematic — Fig 1a/b not yet drawn**]). The backbone is a *frozen* ESM-2 650M encoder⁶; all adaptation is confined to (i) low-rank LoRA updates⁷ (rank 16), (ii) an 8-expert residue mixture-of-experts that specializes over DBD positions, (iii) a contact-aware PWM head that biases motif columns using a DNA-contact signal read from the backbone, and (iv) a learned span gate that commits to the informative motif core. The model is trained on curated protein→PWM pairs (train split `tf_pwm_training_v23`) using DBD-cropped inputs; at inference it requires only the DBD sequence. We report a five-seed ensemble (seeds 42/1/7/13/23) whose members are register-aligned (offset + reverse-complement) to a reference frame before averaging. Full architecture and training details are in Methods.

Sequence-only prediction matches DeepPBS under leakage control

We evaluated TFScope against the structure-based method DeepPBS⁴ on a shared held-out benchmark (Fig. 4). Because naive test/train splits over-report accuracy when homologous TFs leak across the split, we adopted two controls: (i) a 40%-DBD-identity clustered holdout, and (ii) explicit removal of the 20 benchmark genes from DeepPBS’s training for a gene-disjoint retrain. All PWMs are compared with a register-invariant, oracle-aligned motif-content correlation (offset ± 10 , reverse-complement), the fair metric for methods that do not share a fixed coordinate frame.

Under the clustered-holdout benchmark ($n=84$ structures, 41 TFs), the five-seed TFScope ensemble reaches a motif content correlation of **0.664**, improving on the single-seed model (0.629) and every component in the ablation ladder (Fig. 4[**TODO: ladder panel**]); by the mean gate-oracle- r protocol TFScope scores 0.657 versus DeepPBS’s 0.626 (paired difference +0.031, bootstrap $p = 0.30$) — a statistical tie in TFScope’s favor. On the gene-disjoint 20-gene comparison, a DeepPBS ensemble retrained to exclude the test genes scores 0.720 versus TFScope’s 0.685 (paired $\Delta = +0.034$ with a confidence interval crossing zero; the widely cited pretrained DeepPBS value of 0.806 is invalid here because those checkpoints saw all 20 test genes). We emphasize that TFScope attains this parity *without structure*. Per-family accuracy is broken down in Fig. 3, and representative predicted logos in Fig. 2.

An independent structure predictor judges sequence-only motifs as more foldable

To obtain a model-independent quality signal, we asked whether the DNA a model predicts actually forms a confident complex with its TF. For each of 41 TFs we folded the protein with its TFScope-ensemble consensus DNA and, separately, with its DeepPBS consensus DNA, using Boltz-2⁸ with ColabFold⁹ multiple-sequence alignments, and compared the interface confidence (ipTM) of the two complexes (Fig. 5). TFScope’s sequence-only motifs fold at least as confidently as DeepPBS’s structure-derived motifs (mean ipTM **0.957** vs 0.923; mean $\Delta = +0.035$; TFScope favored in **32/41** TFs; Wilcoxon $p = 1 \times 10^{-5}$; complex pLDDT 0.948 vs 0.921). A length control confirms the effect is not an artifact of motif length (Spearman $\rho = -0.43$ between ΔipTM and $\Delta\text{motif-length}$). Because Boltz-2 is independent of both predictors, this provides orthogonal evidence that TFScope’s sequence-only readout is biophysically coherent.

The frozen backbone encodes DNA-contact residues and family identity

Why does a sequence-only model reach structure-level accuracy? We probed the frozen ESM-2 backbone directly. A linear probe on frozen layer-33 residue embeddings predicts which DBD residues contact DNA (within 4.5 Å in co-crystals) with pooled AUROC **0.95** and AUPRC 0.78 across 2,324 complexes (240,072 residues; GroupKFold with no TF shared across folds; Fig. 6). Rendered onto a homeodomain–DNA co-crystal (PBX1, PDB 1B72), the residues TFScope recognizes as DNA-contacting coincide with the recognition-helix positions that insert into the major groove (Fig. 6). Independently, frozen DBD embeddings organize TF families (k-NN purity 0.83; linear family-probe accuracy 0.96), [**TODO: cite ESM DBD family diagnostic figure**]. This structural and taxonomic information is present *before* any TFScope training, and the contact-aware head is what converts it into a calibrated motif.

TFScope reads within-family specificity, not just family identity

A model could reach high aggregate accuracy merely by returning a family-average motif. To test this, we decomposed TFScope’s advantage over a random real-PWM baseline into a family-identity component and a within-family component on the clustered holdout. We find that 41% of the edge is explained by family identity but **59%** reflects genuine within-family specificity (permutation $p = 2 \times 10^{-4}$; Fig. 7[**TODO: decomposition panel**]), showing that TFScope resolves clade-level differences, not only family lookup.

Case studies: motif switching and designed proteins

Single amino-acid substitutions in a DBD can rewire specificity. Using a directional switch score, TFScope reproduces the MyoD1 E-box switch induced by the L122R substitution ($\Delta_{\text{switch}} = +9.17$; Fig. 8), and a structure-based ensemble median agrees on the preferred site (CACGTG). Applied to *de novo* designed DBDs, TFScope recovers the intended CAC-type preference to a graded degree (strongest for DBP6, $r = 0.65$; Fig. 9), demonstrating utility beyond natural proteins. [**PLACEHOLDER — expand designed-DBP paragraph once all design cases finalized**]

Limitations: insensitivity to specificity-altering point mutations

We characterize the model’s principal failure mode transparently. On the Barrera¹⁰ panel of homeodomain wild-type/mutant pairs, TFScope is largely mutation-*blind*: it predicts a mean absolute motif change of 0.005 against an observed 0.180, with near-zero correlation to the measured effect.

This limitation is inherited from the frozen backbone, whose embeddings change little under single substitutions; it coexists with the successful *directional* switch score above because the latter reads a coarse preference shift rather than a fine magnitude. Overcoming point-mutation sensitivity without sacrificing the model’s structure-free generality is the main open problem (Discussion).

Discussion

TFScope shows that accurate TF specificity prediction does not require a protein–DNA structure: a frozen protein language model already carries the recognition signal, and a small structure-free adapter suffices to read it out to motif-level accuracy that matches a strong structure-based method under strict leakage control. Two aspects make the parity credible rather than a metric artifact. First, we control TF leakage explicitly — the dominant reason sequence-only/structure-based comparisons have been unreliable — and still observe parity. Second, an independent structure predictor, blind to both models, finds TFScope’s sequence-only motifs to be at least as foldable as DeepPBS’s.

The interpretability analyses locate the signal: the frozen backbone linearly encodes DNA-contact residues and family identity, and TFScope’s advantage is majority within-family, not family lookup. This reframes what a PLM “knows” about DNA binding and suggests the backbone, not the adapter, is the ceiling. Consistent with that, swapping the motif head for a biophysical recognition-energy decoder did not help ($0.554 < 0.629$), and swapping the backbone to a weaker encoder hurt — the informative bottleneck is the frozen representation itself.

The clearest limitation, mutation-blindness, follows directly: single substitutions barely move frozen embeddings, so fine specificity changes are under-read even though family- and clade-level preferences are captured. Progress likely requires backbone signals that are sensitive to individual residues — for example, contact- or structure-conditioned features — without reintroducing a hard structure dependency at inference. **[PLACEHOLDER — position relative to concurrent PLM-based specificity work once literature search (Phase 1) is complete]**

For practitioners, TFScope provides specificity predictions wherever a DBD sequence is available — including the majority of TFs without co-crystals and the growing space of designed DNA-binding proteins — at the cost of a single forward pass and no structure.

Methods

Model architecture

Frozen ESM-2 650M (`esm2_t33_650M_UR50D`, layer 33) backbone; LoRA adapters (rank 16) on attention projections; an 8-expert residue mixture-of-experts over DBD positions; a contact-aware PWM head (v18 contact bias) that injects a DNA-contact prior derived from the backbone; and a learned span gate committing to the motif core. Output is a $4 \times L$ PWM. **[TODO: exact layer list, LoRA target modules, MoE routing, head equations, param counts]**

Training data and splits

Protein→PWM pairs from `tf_pwm_training_v23` with DBD-cropped inputs (`dbd_start=0`). Held-out evaluation uses a 40%-DBD-identity clustered split and a gene-disjoint 20-gene set for the DeepPBS comparison. **[TODO: data sources (JASPAR/HT-SELEX/CIS-BP), counts per family, dedup/clustering procedure, exact train/val/test sizes]**

Ensembling

Five seeds (42/1/7/13/23). Each member commits its own motif core via the span gate; members are register-aligned to the seed-42 core (offset ± 10 , reverse-complement, minimum overlap 3) and the aligned PWMs are averaged and renormalized.

Evaluation metrics

Panel A: oracle-aligned motif *content* correlation (register-invariant, r). Panel B: end-to-end coverage-aware correlation (covR) with the learned gate. Mean/median gate-oracle- r aligns each method's IC core to ground truth before scoring. **[TODO: precise metric definitions + equations]**

DeepPBS comparison

DeepPBS⁴ run in a dedicated environment (**[TODO: torch/PyG versions]**); `prot_shape` inputs; 5-model ensemble. Gene-disjoint retrain removes the 20 benchmark genes from training. The pretrained checkpoints are leakage-contaminated for this benchmark and reported only as a reference.

Foldability assay

For each of 41 TFs, the protein was folded with (a) its TFScope-ensemble consensus dsDNA and (b) its DeepPBS consensus dsDNA using Boltz-2⁸ (ColabFold MSA server⁹; identical protein per pair, so MSAs are reused). Interface confidence (ipTM) and complex pLDDT were read from Boltz confidence outputs. Length control: Spearman correlation of ΔipTM with $\Delta\text{motif-length}$.

Contact-residue and family probes

Frozen layer-33 residue embeddings; linear probe for DNA-contact residues (4.5 Å cutoff from co-crystals), GroupKFold by protein sequence, 2,324 complexes. Family organization: k-NN purity and linear family probe on frozen DBD embeddings. PyMOL renders write the contact-probe score into the B-factor column of PDB 1B72.

Within-family decomposition

[TODO: describe permutation procedure separating family-identity vs within-family contributions to the edge over a random-real-PWM baseline]

Mutation sensitivity and switch score

Barrera¹⁰ wild-type/mutant homeodomain pairs; mean absolute PWM change vs measured specificity change. Directional switch score: **[TODO: definition of Δ_{switch}]**.

Statistics

Bootstrap and Wilcoxon signed-rank tests as noted; permutation test for the variance decomposition. **[TODO: n, seeds, CI method, multiple-comparison handling]**

Code and data availability

[TODO: repository URL, checkpoint DOI, processed-data deposition]

Declarations

Author contributions (CRediT). [TODO: fill per author]

Competing interests. The authors declare no competing interests. [TODO: confirm]

Funding. [TODO: grants]

Data availability. [TODO: deposition + accession]

Code availability. [TODO: repo + license]

AI-use statement. [TODO: per Nature Methods policy; drafting assistance used]

Figures

[PLACEHOLDER — Fig 1a/b architecture schematic not yet drawn]

Figure 1: TFSScope architecture. Frozen ESM-2 650M backbone with LoRA, residue mixture-of-experts, contact-aware PWM head, and span gate mapping a DBD sequence to a PWM. [TODO: draw schematic]

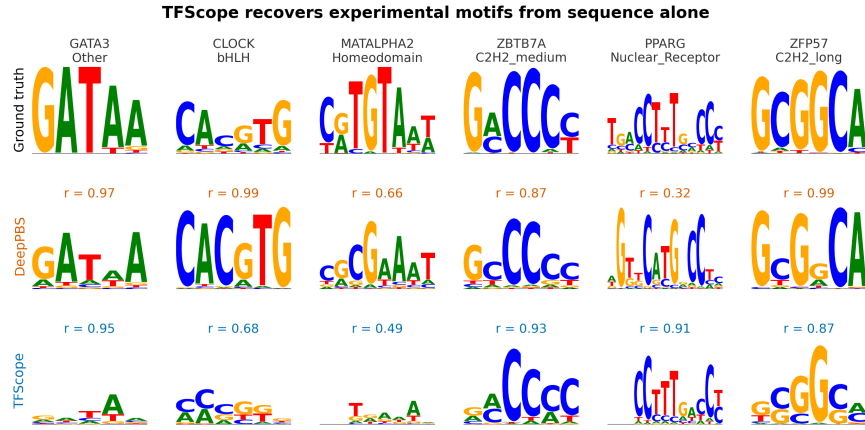


Figure 2: Representative predicted motifs. TFSScope-ensemble predicted logos for held-out TFs.

v24-ensemble: benchmark recovery by family (deeppbs_cluster40 test, 84 samples / 41 TFs)

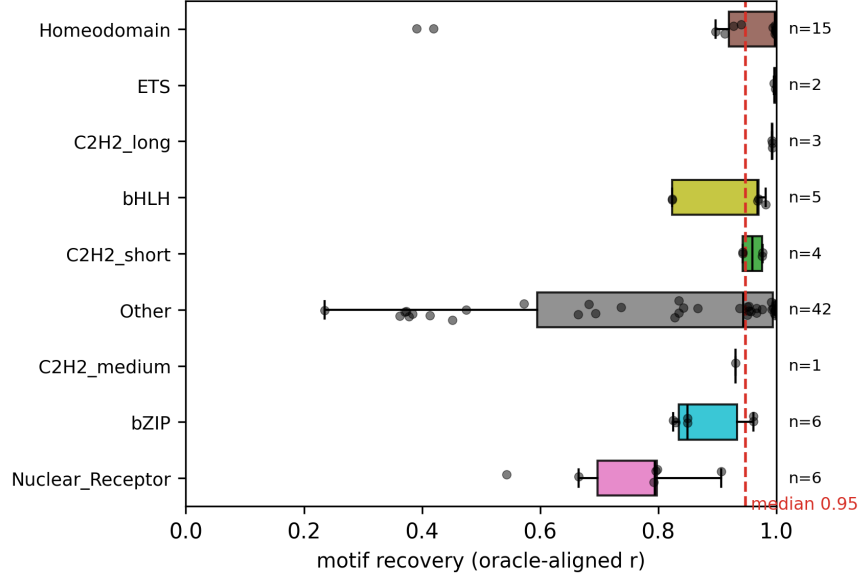


Figure 3: Leakage-controlled benchmark, per family. Motif content correlation on the clustered holdout ($n=84$, 41 TFs); ensemble 0.664. Head-to-head with DeepPBS is a statistical tie (mean gate-oracle r 0.657 vs 0.626; bootstrap $p = 0.30$).

[PLACEHOLDER — benchmark bar + ablation ladder panel]

Figure 4: Benchmark and ablation ladder. [TODO: assemble from results/iclr_phase1_apples_to_apples].

v24 ensemble: sequence-only consensus vs structure-based DeepPBS, judged by Boltz-2 foldability

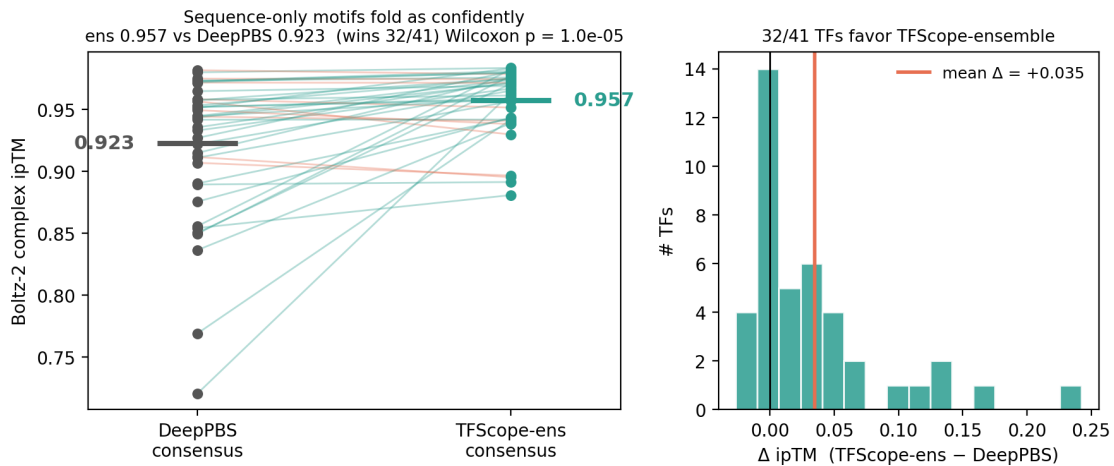


Figure 5: Independent foldability. Boltz-2 complex ipTM for TFScope-ensemble vs DeepPBS consensus DNA (41 TFs): 0.957 vs 0.923, 32/41 favor TFScope, Wilcoxon $p = 1 \times 10^{-5}$.

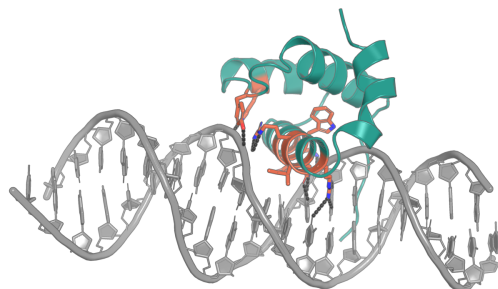


Figure 6: DNA-contact residues recognized by TFScope. PBX1 homeodomain (PDB 1B72): recognition-helix residues scored as DNA-contacting by the frozen- backbone contact probe (AUROC 0.95) shown as orange sticks with H-bonds to the major-groove bases.

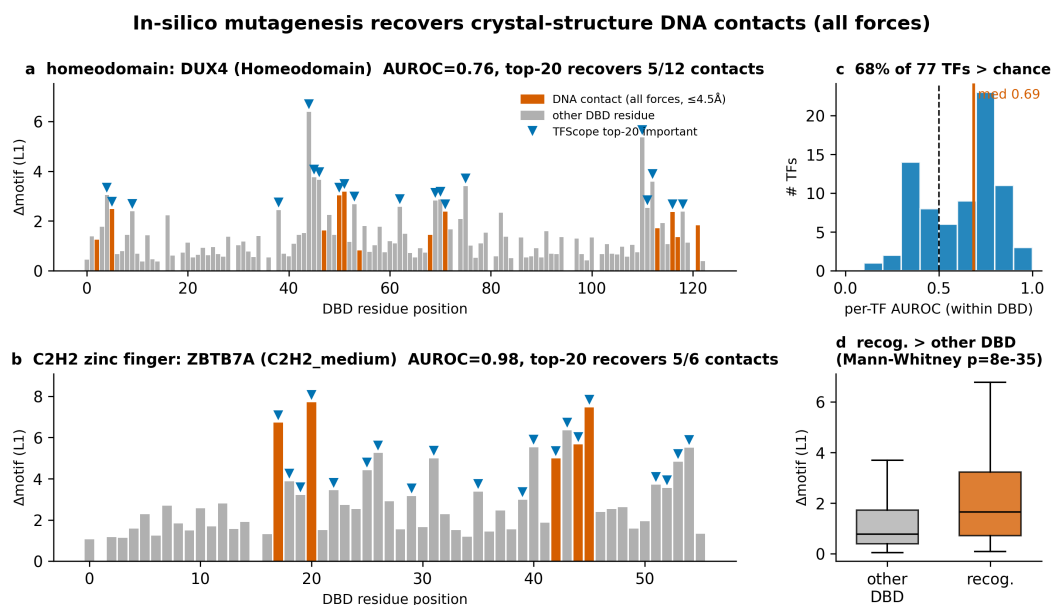


Figure 7: In-silico mutagenesis / within-family signal. [TODO: finalize caption + add decomposition panel].

L122R(v24) shifts MyoD1's predicted preference toward the MYC-like E-box (CACGTG)

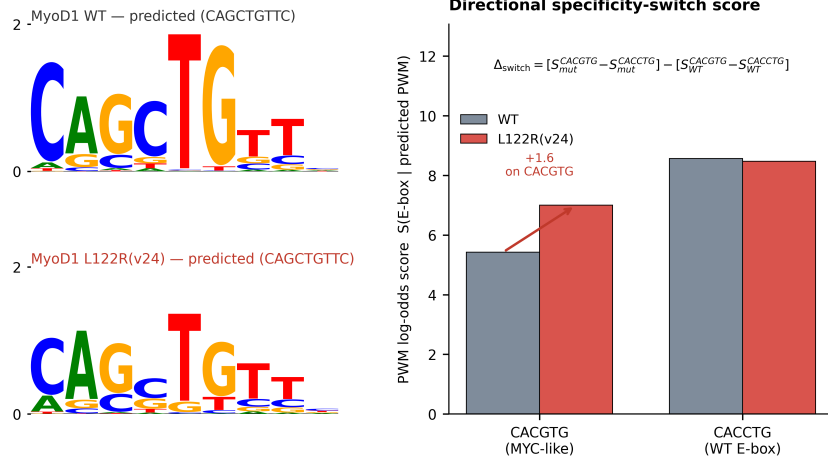


Figure 8: Single-mutation motif switch. MyoD1 L122R E-box switch, $\Delta_{\text{switch}} = +9.17$.

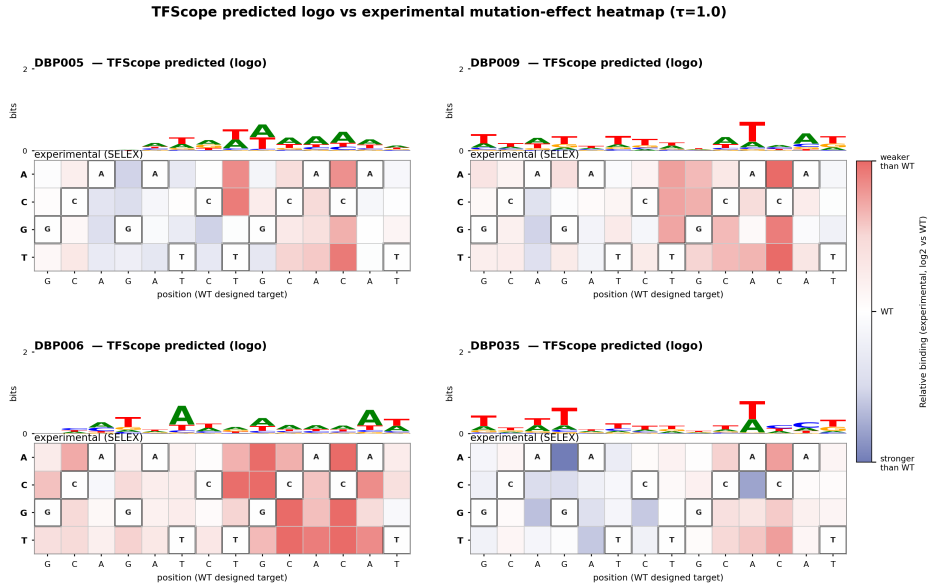


Figure 9: Designed DNA-binding proteins. TFScope recovers intended CAC- type preference (strongest DBP6, $r = 0.65$).

References

- [1] Matthew T. Weirauch, Ally Yang, Mihai Albu, et al. Determination and inference of eukaryotic transcription factor sequence specificity. *Cell*, 158(6):1431–1443, 2014. doi: 10.1016/j.cell.2014.08.009.
- [2] Jaime A. Castro-Mondragon, Rafael Riudavets-Puig, Ieva Rauluseviciute, et al. Jaspar 2022: the 9th release of the open-access database of transcription factor binding profiles. *Nucleic Acids Research*, 50(D1):D165–D173, 2022. doi: 10.1093/nar/gkab1113.
- [3] Arttu Jolma, Jian Yan, Thomas Whittington, et al. Dna-binding specificities of human transcription factors. *Cell*, 152(1-2):327–339, 2013. doi: 10.1016/j.cell.2012.12.009.
- [4] Raktim Mitra, Alan M. Cohen, Jared M. Sagendorf, Helen M. Berman, and Remo Rohs. Geometric deep learning of protein–dna binding specificity. *Nature Methods*, 2024. DeepPBS. **[TODO: verify volume/pages/DOI]**.
- [5] Alexander Rives, Joshua Meier, Tom Sercu, Siddharth Goyal, Zeming Lin, Jason Liu, Demi Guo, Myle Ott, C. Lawrence Zitnick, Jerry Ma, and Rob Fergus. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proceedings of the National Academy of Sciences*, 118(15):e2016239118, 2021. doi: 10.1073/pnas.2016239118.
- [6] Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, Allan dos Santos Costa, Maryam Fazel-Zarandi, Tom Sercu, Salvatore Candido, and Alexander Rives. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023. doi: 10.1126/science.ade2574.
- [7] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.
- [8] Jeremy Wohlwend, Gabriele Corso, Saro Passaro, et al. Boltz-1: Democratizing biomolecular interaction modeling. *bioRxiv*, 2024. **[TODO: update to Boltz-2 citation + DOI]**.
- [9] Milot Mirdita, Konstantin Schütze, Yoshitaka Moriwaki, Lim Heo, Sergey Ovchinnikov, and Martin Steinegger. Colabfold: making protein folding accessible to all. *Nature Methods*, 19: 679–682, 2022. doi: 10.1038/s41592-022-01488-1.
- [10] Luis A. Barrera, Anastasia Vedenko, Jesse V. Kurland, et al. Survey of variation in human transcription factors reveals prevalent dna binding changes. *Science*, 351(6280):1450–1454, 2016. doi: 10.1126/science.aad2257.